

Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems

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International Symposium on Forecasting
June 28 – July 1 2026, Montreal, Canada



Outline

Accuracy-Stability Trade-off

Experimental Design

Empirical Results

Conclusions

Accuracy-Stability Trade-off

What is Forecast Stability?

Horizontal stability

- ▶ Measures how forecasts change across *different future time point* from the *same origin*.

Vertical stability

- ▶ Measures how forecasts change for the *same future time point* when the model is re-estimated at *different origins*.

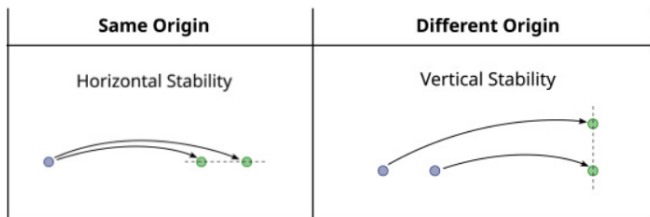


Figure 1: Godahewa, R., et al. (2025). *On forecast stability*.

Why it matters?

Forecasting drives operational decisions (inventory planning, workforce scheduling, budgeting and supply chain decisions, etc.), and ...

unstable forecasts lead to:

- ▶ revisions to plans and decisions,
- ▶ planning inefficiencies and higher costs,
- ▶ loss of credibility and trust in the forecasting system.

Why can a trade-off arise?

- ▶ Forecasting models are updated frequently.
- ▶ More frequent retraining $\Rightarrow$ more accurate forecasts.
- ▶ Frequent model updates may produce structural model changes $\Rightarrow$ forecasts can shift.

Key questions

- ▶ Can we maintain high accuracy while producing stable forecasts?
- ▶ Is frequent retraining necessary?

Experimental Design

Datasets & Models

Dataset	Frequency	N. Series	Test Set	Horizon
M5	Daily	30,490	364	28

Machine Learning	Deep Learning
Linear Regression (LR)	MLP
XGBoost	TCN
LightGBM (LGBM)	NBEATSx
CatBoost	NHITS

Evaluation Strategy: Rolling Origin

- ▶ **Why?** Out-of-sample testing assesses generalization
- ▶ **How?**
 - ▶ Train/test split, preserving temporal order
 - ▶ Train model on initial training set
 - ▶ Forecast h steps ahead
 - ▶ Shift origin forward by p steps
 - ▶ Repeat until the test set is exhausted
- ▶ **Window types:**
 - ▶ Fixed window: Only the most recent n observations used
 - ▶ Expanding window: Include all historical data up to the current origin
- ▶ **In our study:** Expanding window with a step size $p = 1$

Evaluation Metrics

Accuracy

- ▶ **Point:** Root Mean Squared Scaled Error (RMSSE) (1)
- ▶ **Probabilistic:** Scaled Multi-Quantile Loss (SMQL) (2),(3)

Stability

- ▶ **Point:** Symmetric Mean Absolute Percentage Change (sMAPC) (4)
- ▶ **Probabilistic:** Symmetric Multi-Quantile Percentage Change (sMQPC) (5),(6)

Sustainability Metrics

- ▶ Computing Time (CT)
- ▶ Cost (\$) and Savings (%)

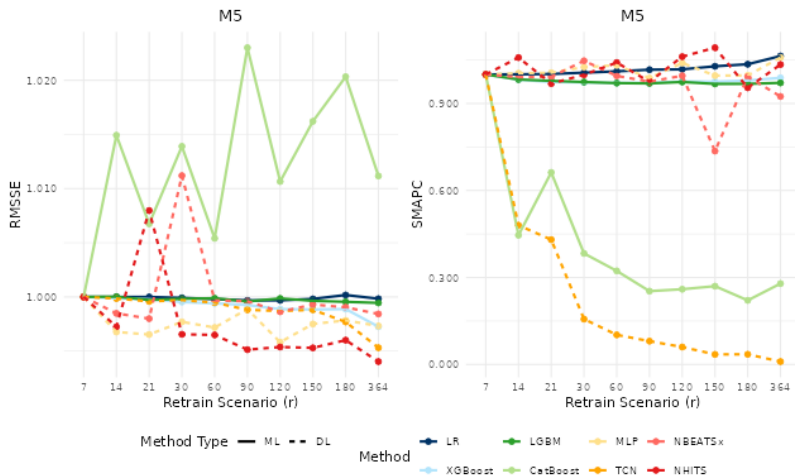
Simulation of the savings of each retraining scenario for a large retailer (200,000 SKUs and 5,000 stores), assuming some standard costs for computing services (\$3.5/hour).

Retraining Scenarios

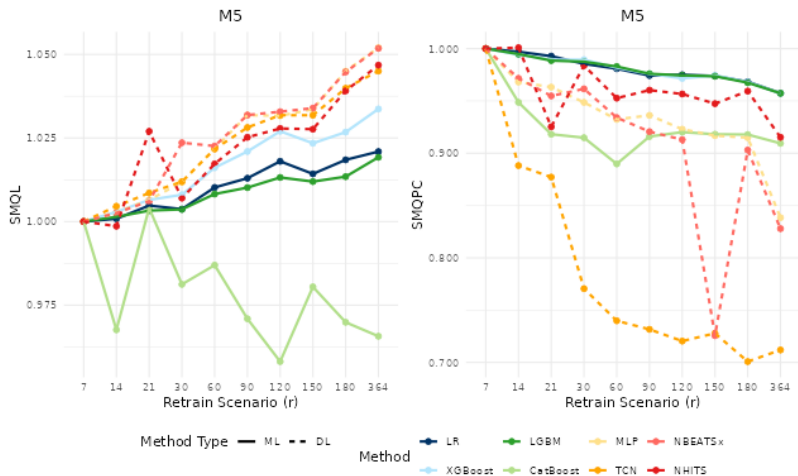
- ▶ **Retrain scenario (r):** the interval at which the model is re-trained or updated.
- ▶ **Scenarios based on data frequency:**
 - ▶ Daily: $r = \{7, 14, 21, 30, 60, 90, 120, 150, 180, 364\}$
- ▶ **Three strategies:**
 - ▶ Continuous retraining, $r = 7$, baseline scenario
 - ▶ No retraining, $r = 364$
 - ▶ Periodic retraining, $7 < r < 364$

Empirical Results

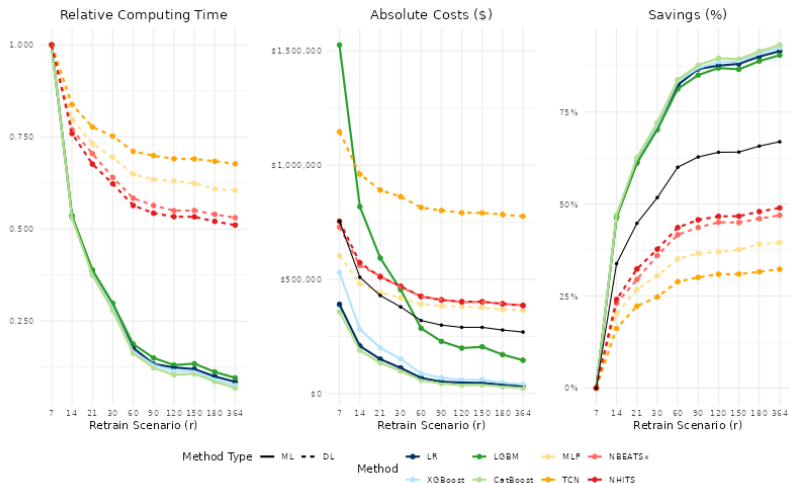
Point Forecasting



Probabilistic Forecasting



Sustainability



Conclusions

Key Takeaways

- ▶ Highly frequent retraining may not be necessary
- ▶ Periodic retraining strategies allow for balancing accuracy and stability
- ▶ Reducing the retraining frequency supports more sustainable forecasting systems (lower costs of producing forecasts)

Future Research

- ▶ Retraining under concept drifts / structural breaks

References

Zanotti, M. (2026). *Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems*. arXiv. (url)

Zanotti, M. (2025). *On the retraining frequency of global models in retail demand forecasting*. Machine Learning with Applications, 22, 100769. (url)

Thank you!

Appendix

Accuracy Metrics - Math

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{h} \sum_{t=n+1}^{n+h} (y_t - \hat{y}_t)^2}{\frac{1}{n-s} \sum_{t=s+1}^n (y_t - y_{t-s})^2}} \quad (1)$$

$$\text{SQL}(\alpha) = \frac{\frac{1}{h} \sum_{t=n+1}^{n+h} \left(\alpha \cdot (y_t - \hat{q}_t^{(\alpha)}) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} \leq y_t} + (1-\alpha) \cdot (\hat{q}_t^{(\alpha)} - y_t) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} > y_t} \right)}{\frac{1}{n-s} \sum_{t=s+1}^n |y_t - y_{t-s}|} \quad (2)$$

$$\text{SMQL} = \frac{1}{|Q|} \sum_{\alpha \in Q} \text{SQL}(\alpha) \quad (3)$$

Stability Metrics - Math

$$\text{sMAPC} = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{y}_{t,n} - \hat{y}_{t,n-1}|}{|\hat{y}_{t,n}| - |\hat{y}_{t,n-1}|} \quad (4)$$

$$\text{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|}. \quad (5)$$

$$\text{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{sQPC}(\alpha), \quad (6)$$

Symmetric Multi-Quantile Percentage Change

The idea of sQPC is to mirror the sMAPC, but using quantile forecasts at different origins for some specific level α .

$$\text{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|} \quad (7)$$

The sMQPC simply averages the SQPC across the set of quantiles $\mathcal{Q}$, measuring overall probabilistic instability.

$$\text{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{sQPC}(\alpha), \quad (8)$$

Computing Setup

- ▶ Microsoft Azure NC6s v3 VM
- ▶ 6 vCPUs, 1 GPU, 112 GB RAM
- ▶ Nixtla's libraries: `statsforecast`, `mlforecast`, `neuralforecast`